```
!pip install torch transformers datasets peft accelerate -q
```

```
import torch
from transformers import AutoModelForSequenceClassification, AutoTokenizer
from datasets import load_dataset
from peft import LoraConfig, get_peft_model
```

```
model_name = "distilbert-base-uncased"

tokenizer = AutoTokenizer.from_pretrained(model_name)

model = AutoModelForSequenceClassification.from_pretrained(
    model_name,
    num_labels=2
)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingfac
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datas
  warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable h
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hu
```

config.json: 100% 483/483 [00:00<00:00, 5.88kB/s]

tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 427B/s]

vocab.txt: 232k/? [00:00<00:00, 1.69MB/s]

tokenizer.json: 466k/? [00:00<00:00, 3.26MB/s]

model.safetensors: 100% 268M/268M [00:04<00:00, 176MB/s]

Loading weights: 100% 100/100 [00:00<00:00, 117.57it/s, Materializing param=distilbert.transformer.layer.5.sa_layer_norm.w

```
DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                      | Status     |
-------------------------+------------+-
vocab_projector.bias     | UNEXPECTED |
vocab_layer_norm.bias    | UNEXPECTED |
vocab_layer_norm.weight  | UNEXPECTED |
vocab_transform.bias     | UNEXPECTED |
vocab_transform.weight   | UNEXPECTED |
pre_classifier.bias      | MISSING    |
pre_classifier.weight    | MISSING    |
classifier.bias          | MISSING    |
classifier.weight        | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expec
- MISSING   :those params were newly initialized because missing from the checkpoint. Consider tra
```

```
dataset = load_dataset("imdb")
```

```
README.md:          7.81k/?  [00:00<00:00,  130kB/s]
plain_text/train-00000-of-00001.parquet:  100%  21.0M/21.0M  [00:00<00:00,  104MB/s]
plain_text/test-00000-of-00001.parquet:  100%  20.5M/20.5M  [00:00<00:00,  102MB/s]
plain_text/unsupervised-00000-of-00001.p(…):  100%  42.0M/42.0M  [00:00<00:00,  108MB/s]
Generating train split:  100%  25000/25000  [00:01<00:00,  29411.92 examples/s]
Generating test split:  100%  25000/25000  [00:01<00:00,  24790.30 examples/s]
Generating unsupervised split:  100%  50000/50000  [00:01<00:00,  36509.70 examples/s]
```

```
def tokenize_function(example):
    return tokenizer(example["text"], truncation=True, padding="max_length")

tokenized_dataset = dataset.map(tokenize_function, batched=True)
```

```
Map:  100%  25000/25000  [01:01<00:00,  393.99 examples/s]
Map:  100%  25000/25000  [00:52<00:00,  678.01 examples/s]
Map:  100%  50000/50000  [01:04<00:00,  819.16 examples/s]
```

```
train_dataset = tokenized_dataset["train"].shuffle(seed=42).select(range(500))
test_dataset = tokenized_dataset["test"].shuffle(seed=42).select(range(200))
```

```
lora_config = LoraConfig(
    r=8,
    lora_alpha=16,
    target_modules=["q_lin", "v_lin"],
    lora_dropout=0.1,
    bias="none",
    task_type="SEQ_CLS"
)
```

```
peft_model = get_peft_model(model, lora_config)

peft_model.print_trainable_parameters()
```

```
from transformers import TrainingArguments, Trainer

training_args = TrainingArguments(
    output_dir="./results",
    learning_rate=2e-4,
    per_device_train_batch_size=8,
    num_train_epochs=1,
    logging_steps=10,
    save_strategy="no"
)

trainer = Trainer(
    model=peft_model,
    args=training_args,
    train_dataset=train_dataset
)

trainer.train()
```

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memo
  super().__init__(loader)
```

[46/63 16:05 < 06:13, 0.05 it/s, Epoch 0.71/1]

Step	Training Loss
10	0.707668
20	0.668999
30	0.669488
40	0.635658

[63/63 22:28, Epoch 1/1]

Step	Training Loss
10	0.707668
20	0.668999
30	0.669488
40	0.635658
50	0.636174
60	0.624832

```
TrainOutput(global_step=63, training_loss=0.6552340095005338, metrics={'train_runtime':
1381.6504, 'train_samples_per_second': 0.362, 'train_steps_per_second': 0.046, 'total_flos':
67369703424000.0, 'train_loss': 0.6552340095005338, 'epoch': 1.0})
```

```
trainer.evaluate(test_dataset)
```

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memo
  super().__init__(loader)
```

[25/25 03:00]

```
{'eval_loss': 0.6240664124488831,
 'eval_runtime': 188.6364,
 'eval_samples_per_second': 1.06,
 'eval_steps_per_second': 0.133,
 'epoch': 1.0}
```

```
text = "The movie was fantastic and inspiring!"

inputs = tokenizer(text, return_tensors="pt")

outputs = peft_model(**inputs)

prediction = torch.argmax(outputs.logits, dim=1)

print("Predicted label:", prediction.item())
```

```
Predicted label: 1
```

Start coding or generate with AI.